

Technical Document #461: Data Engineering - Comprehensive Overview

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Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Partitioning divides data into smaller, more manageable pieces.
- Data lakes store raw data in its native format.
- Schema evolution handles changes in data structure over time.